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Roll No

MI-801 (GS)

B.E. VIII Semester Examination, June 2020

Grading System (GS)

Mining Surveying - III

Time : Three Hours

Maximum Marks : 70

Note : i) Attempt any five questions.

ii) All questions carry equal marks.

1. With the help of a neat diagram explain the Co-Planning method. Also list out the precautions to be taken into account during the survey.
2. What is the application of vertical and oblique photographs?
3. The altitude of two proposed station A and B, 100 km apart, are respectively 420m and 700m. The intervening obstruction situated at C 70 km from A has an elevation of 478m. As certain if A and B are indivisible and if necessary. Find by how much B should be raised so that the line of sight must nowhere be less than 3m above the surface of the ground.
4. What are the systems of co-ordinates employed to locate position of a heavenly body? Why it is necessary to have several systems instead of one.
5. Determine the hour angle and declination of a star from the following data
 - i) Altitude of the star = $28^{\circ} 24'$
 - ii) Azimuth of the star = $50^{\circ} W$
 - iii) Latitude of the place of observation = $48^{\circ} 30' N$
6. Calculate the sag correction for a 30 m steel tape under a pull of 10 kg in three equal spans of 10 cm each weight of 1 cubic cm of steel = 7.86 gm. Area of cross-section of the tape = 0.08 sq.cm.
OR
What do you understand by triangulation survey? Give its classification with specifications.
7. Why correlation survey is necessary in mines? How it can be done when only one vertical shaft is available for correlation?
8. Explain the following with neat sketch:(Any four)
 - a) Zenith
 - b) Nadir
 - c) Celestial equation
 - d) The observers meridian
 - e) The visible horizon
 - f) Declination
 - g) Azimuth
